Jiaru Li

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Research Interest

Theoretical and computational astrophysics: nonlinear disk hydrodynamics, orbital dynamics, and planet-disk interactions. Applications to exoplanets, black holes, accretion processes and distorted accretion disks, the dynamical formation of gravitational-wave sources, and black-hole dynamics in active galactic nuclei (AGN). Methods include analytical modeling, hydrodynamic simulations, and N-body integrations. Additional interests include machine learning and large language models (LLMs).

Academic Appointments

Postdoctoral Research Associate *2026–present*
Department of Applied Mathematics and Theoretical Physics, University of Cambridge, UK
Supervisor: Prof. Roman Rafikov

CIERA Postdoctoral Fellow *2023–2026*
Center for Interdisciplinary Exploration and Research in Astrophysics, Northwestern University, USA
Independent Fellowship

CSES Graduate Student Fellow *2020–2022*
Theoretical Division, Los Alamos National Laboratory, USA
Mentor: Dr. Hui Li

Graduate Research Assistant *2017–2023*
Department of Astronomy, Cornell University, USA
Advisor: Prof. Dong Lai

Undergraduate Research Assistant *2015–2017*
Department of Physical and Environmental Sciences, University of Toronto Scarborough, Canada
Supervisor: Prof. Artur Izmaylov

Education

Cornell University *2017–2023*
Ph.D. in Astronomy and Space Sciences (May 2023).
Dissertation: *Dynamical Evolution of Multi-Orbiter Systems: Application to Planets, Protoplanetary Disks, and Black Holes.*
Advisor: Prof. Dong Lai
M.S. in Astronomy (December 2019).

University of Toronto Scarborough *2013–2017*
Honours B.Sc. in Physics (Specialist) and Mathematics (Major).

Selected Honors and Awards

2023 CIERA Postdoctoral Fellowship, Northwestern University.
2020 CSES Student Fellowship, Los Alamos National Laboratory.
2017 New Graduate Student Fellowship, Cornell University.
2017 Governor General's Silver Medal Nomination, University of Toronto Scarborough.
2017 Samuel Beatty In-Course Scholarship, University of Toronto.
2016 University of Toronto Excellence Award.
2014 A. D. Allen Memorial Scholarship, University of Toronto at Scarborough

Research Mentorship and Teaching

Students Mentored:

- **Jupiter Ding** (2025–2026), graduate student, Northwestern University; protoplanetary-disk research.
- **Jesse Richter** (2024–2026), undergraduate student, Northwestern University.
- **Dieran Wang** (2024–2026), undergraduate student at Shandong University, subsequently a Ph.D. student at Shanghai Jiao Tong University; lead author of a 2026 *ApJ* paper.
- **Kecheng Qian** (2023–2024), undergraduate researcher at Cornell University, subsequently a Ph.D. student at UC Berkeley; lead author of a 2024 *ApJ* paper.

Teaching assistant:

- **Cornell University** (2018–2019): *Our Solar System*; *From New Worlds to Black Holes*.
- **University of Toronto** (2014-2017): Calculus I–II; Physics I–II; Linear Algebra II; Discrete Mathematics.

Professional Service

- Referee for *The Astrophysical Journal*, *The Astrophysical Journal Letters*, *Monthly Notices of the Royal Astronomical Society*, and *Astronomy & Astrophysics*, 2023–present.
- Proposal-review panelist, NASA Exoplanets Research Program (XRP), 2024.
- Scientific Organizing Committee, *CIERA Fellows at 15* conference, 2025.
- Organizer, CIERA Journal Club and summer arXiv discussions, 2024–2025.
- Session chair, AAS Division on Dynamical Astronomy (DDA) meeting, 2024; CIERA Fellows conference, 2025.

Selected Invited Presentations

- 2026 Jan — Department of Astronomy Colloquium, Westlake University, China.
- 2025 Dec — International Conference on Exoplanets and Planet Formation, Shanghai, China.
- 2024 Oct — Theoretical Astrophysics Center Seminar, UC Berkeley, USA.
- 2024 Oct — Transient Phenomena and Physical Processes Around Supermassive Black Holes, Tsung-Dao Lee Institute, China.
- 2024 Sep — Astronomy Tea Talk, Indiana University Bloomington, USA.
- 2024 Aug — New Ideas on the Origin of Black Hole Mergers, Niels Bohr Institute, Denmark.
- 2024 Mar — Center for Theory and Computation Seminar, University of Maryland, USA.
- 2024 Jan — ET Science Seminar, Shanghai Astronomical Observatory, China.
- 2023 Dec — Texas Symposium on Relativistic Astrophysics, Shanghai, China.
- 2023 Mar — AGN Santa Fe: *Where Are the Objects in AGN Disks?*, USA.
- 2022 Nov — Center for Exoplanets and Habitable Worlds Seminar, Penn State, USA.
- 2022 Oct — Center for Relativistic Astrophysics Seminar, Georgia Tech, USA.

Publications

Publication overview. 17 published or accepted papers (including 9 first-authored + 2 student-led papers); 6 additional manuscripts submitted or in preparation (October 2026). 450+ citations as of October 2026. **Jiaru Li's name is bolded.** An asterisk (*) denotes a student mentored by Jiaru Li.

Astrophysics: published papers

1. **Jiaru Li**, O'Connor, & Rasio (2026), "Intruder Alert: Breaking resonant chains with planetesimal flybys," *ApJL*, 998, L5.
2. Wang*, **Jiaru Li**, & Lai (2026), "Spin and Obliquity Distributions of Low-mass Planets Shaped by Dynamical Instability," *ApJ*, 996, 3.
3. **Jiaru Li**, Rodet, & Lai (2024), "Dynamical instability in multi-orbiter systems with gas friction," *MNRAS*, 528,

1198.

4. Qian*, **Jiaru Li**, & Lai (2024), "Dynamical Friction Models for Black Hole Binary Formation in Active Galactic Nucleus Disks," *ApJ*, 962, 143.
5. **Jiaru Li** & Lai (2023), "Resonant Excitation of Planetary Eccentricity due to a Dispersing Eccentric Protoplanetary Disk: A New Mechanism of Generating Large Planetary Eccentricities," *ApJ*, 956, 17.
6. **Jiaru Li**, Dempsey, Li, Lai, & Li (2023), "Hydrodynamical Simulations of Black Hole Binary Formation in AGN Disks," *ApJL*, 944, L42.
7. **Jiaru Li**, Lai, & Rodet (2022), "Long-term Evolution of Tightly Packed Stellar Black Holes in AGN Disks: Formation of Merging Black Hole Binaries via Close Encounters," *ApJ*, 934, 154.
8. Li, Dempsey, Li, Li, & **Jiaru Li** (2022), "Hot Circumsingle Disks Drive Binary Black Hole Mergers in Active Galactic Nucleus Disks," *ApJL*, 928, L19.
9. Li, Dempsey, Li, Li, & **Jiaru Li** (2021), "Orbital Evolution of Binary Black Holes in Active Galactic Nucleus Disks: A Disk Channel for Binary Black Hole Mergers?" *ApJ*, 911, 124.
10. **Jiaru Li**, Dempsey, Li, & Li (2021), "Ring Formation in Protoplanetary Disks Driven by an Eccentric Instability," *ApJ*, 910, 79.
11. **Jiaru Li**, Lai, Anderson, & Pu (2021), "Giant planet scatterings and collisions: hydrodynamics, merger-ejection branching ratio, and properties of the remnants," *MNRAS*, 501, 1621.
12. **Jiaru Li** & Lai (2020), "Planetary Spin and Obliquity from Mergers," *ApJL*, 898, L20.

Astrophysics: ongoing manuscript

- **Jiaru Li** & Lithwick (2026), "Bending, Bouncing, and Breaking of Warped Accretion Disks," to be submitted. [Draft PDF].
- **Jiaru Li** & Lai (2026), "Prograde and Retrograde Black Hole Binary Formation in AGN Disks," manuscript in preparation.

Computer science and machine learning

1. Huang, Zhang, Cai, Niu, **Jiaru Li**, Wang, Zhang, Jiang, Zhao, Penn, and Mirzasoleiman, "HR-Merge: Model Merging with Headroom-Weighted Residual Refinement," *NeuralArtifacts Workshop 2026*.
2. Lu, **Jiaru Li** (co-first author), Zhao, Chen, Wang, Cai, Li, & He (2026), "On the Representation Geometry of LoRA Model Merging," *ACL 2026*.
3. Zhao, Zhang, **Jiaru Li**, Niu, He, Min, & Penn, "Tiny Budgets, Big Gains: Parameter Placement Strategy in Parameter Super-Efficient Fine-Tuning," *EMNLP 2025*.
4. Zhao, Zhang, Huang, Mao, Niu, Chen, Wang, Yang, Ding, Cai, Liu, Penn, and **Jiaru Li**, "TARE: Timescale-Aware Row-wise Energy Redistribution for Muon Optimizers," submitted to ACL, October 2026.
5. Zhao, Wang, Mao, Li, Wu, Chen, **Jiaru Li**, Ding, Penn, and He, "From Visual Artifacts to Forensic Language: A Mechanistic Study of VLM-Based AI-Generated Image Detection," submitted to ACL, October 2026.
6. Cao, **Jiaru Li**, Zhao, Zhang, Sun, and Penn, "ConstellationBench: Evaluating Astronomical Reasoning and Spatial Intelligence in Vision-Language Models," submitted to ACL, October 2026.

Quantum physics and chemical physics

1. **Jiaru Li**, Joubert-Doriol, & Izmaylov (2017), "Geometric phase effects in excited state dynamics through a conical intersection in large molecules: N-dimensional linear vibronic coupling model study," *J. Chem. Phys.*, 147, 064106.
2. Izmaylov, **Jiaru Li**, & Joubert-Doriol (2016), "Diabatic Definition of Geometric Phase Effects," *J. Chem. Theory Comput.*, 12, 5278–5283.